# Curriculum Vitae Europass



## Personal information


| | |
|---|---|
| Surname(s) / First name(s) | Stanculescu Marilena |
| Address(es) | Splaiul Independentei nr. 313, corp EA<br>060042, Bucharest, ROMANIA |
| E-mail(s) | marilena.stanculescu@upb.ro |
| Nationality(-ies) | Romanian |
| Date of birth | 29/03/1977 |
| Gender | F |


## Work experience

| | |
|---|---|
| Dates | Oct 2023 → present |
| Occupation or position held | Professor |
| Main activities and responsibilities | Course Fundamentals of Electrical Engineering I, II – taught in English (Faculty FILS, year II)<br>Course Fundamentals of Electrical Engineering III – taught in English (Faculty FILS, year II)<br>Course Fundamentals of Electrical Engineering – taught in Romanian (A&C, year I)<br>Course Fundamentals of Electrical and Electronic Engineering (FICBI, year II)<br>Course Electromagnetic Control of Product Quality (IE, year IV)<br>Seminar/Laboratory: Fundamentals of Electrical Engineering I, Fundamentals of Electrical Engineering II/Electrical Engineering I (A&C, FILS, ETTI, year I)<br>Member of the PhD student supervision committee<br>PhD supervisor |
| Name and address of employer | National University of Science and Technology POLITEHNICA Bucharest<br>National University of Science and Technology POLITEHNICA Bucharest, Faculty of Electrical Engineering, Splaiul Independentei, no.313, 060042 Bucharest (Romania) |
| Type of business or sector | Education |

| | |
|---|---|
| Dates | 09/2019 → Oct 2023 |
| Occupation or position held | Associate Professor |
| Main activities and responsibilities | Course Fundamentals of Electrical Engineering I, II – taught in English (Faculty FILS, year II)<br>Course Fundamentals of Electrical Engineering III – taught in English (Faculty FILS, year II)<br>Course Fundamentals of Electrical Engineering – taught in Romanian (A&C, year I)<br>Course Fundamentals of Electrical and Electronic Engineering (FICBI, year II)<br>Course Electromagnetic Control of Product Quality (IE, year IV)<br>Seminar/Laboratory: Fundamentals of Electrical Engineering I, Fundamentals of Electrical Engineering II/Electrical Engineering I (A&C, FILS, ETTI, year I)<br>Laboratory Medical Informatics (FIM, Master TIM)<br>Member of the PhD student supervision committee<br>PhD supervisor |
| Name and address of employer | Politehnica University of Bucharest, Faculty of Electrical Engineering<br>Splaiul Independentei, no.313, 060042 Bucharest (Romania) |
| Type of business or sector | Education |


For more information on Europass go to http://europass.cedefop.eu.int
© European Union, 2002-2010 24082010

| | |
|---|---|
| Dates | 02/2012 → 2019 |
| Occupation or position held | Lecturer |
| Main activities and responsibilities | Course Fundamentals of Electrical Engineering I, II – taught in English (Faculty FILS, year II)<br>Course Fundamentals of Electrical and Electronic Engineering (FICBI, year II)<br>Course Electromagnetic Control of Product Quality (IE, year IV)<br>Seminar/Laboratory: Fundamentals of Electrical Engineering I, Fundamentals of Electrical Engineering II/Electrical Engineering I (A&C, FILS, ETTI, year I)<br>Laboratory Professional Communication Techniques (IE, year I) |
| Name and address of employer | Politehnica University of Bucharest, Faculty of Electrical Engineering<br>Splaiul Independentei, no.313, 060042 Bucharest (Romania) |
| Type of business or sector | Education |
| Dates | 02/2009 - 02/2012 |
| Occupation or position held | Teaching Assistant |
| Main activities and responsibilities | Seminars in Fundamentals of Electrical Engineering I, Fundamentals of Electrical Engineering II<br>Laboratory Applied Informatics, Laboratory of Fundamentals of Electrical Engineering. |
| Name and address of employer | Politehnica University of Bucharest, Faculty of Electrical Engineering<br>Splaiul Independentei, no. 313, 060042 Bucharest (Romania) |
| Type of business or sector | Education |
| Dates | 2008 - 02/2009 |
| Occupation or position held | Administrative Director |
| Main activities and responsibilities | Administration |
| Name and address of employer | CRRN Gura Ocnitei<br>Main St., no. 2, Gura Ocnitei (Romania) |
| Type of business or sector | Health |
| Dates | 1998 - 1999 |
| Occupation or position held | Press Officer |
| Main activities and responsibilities | PR<br>Drafting press materials for participants at the Bucharest Auto Show.<br>Bucharest Auto Show, Press Center. |
| Name and address of employer | SAB Bucharest<br>(Romania) |
| Type of business or sector | PR, press |
| Dates | 1997 - 1999 |
| Occupation or position held | Computer Operator |
| Main activities and responsibilities | Database operation |
| Name and address of employer | Trans Romania Trading SRL<br>Bucharest (Romania) |
| Type of business or sector | Trade |
| Dates | 1997 - 2002 |
| Occupation or position held | Tour guide - communication in English |
| Main activities and responsibilities | Development of tourist itineraries |
| Name and address of employer | Travcom International Travel Agency<br>Bucharest (Romania) |
| Type of business or sector | Tourism |
| Dates | 2010 → 2013 |


For more information on Europass go to http://europass.cedefop.eu.int
© European Union, 2002-2010 24082010

| | |
|---|---|
| Occupation or position held | postdoctoral researcher - EXCEL ID 62557 |
| Main activities and responsibilities | Research |
| Name and address of employer | Politehnica University of Bucharest<br>Splaiul Independentei, no. 313, 060042 Bucharest (Romania) |
| Type of business or sector | Education, Research |

## Education and training

| | |
|---|---|
| Dates | 07.10.2021 |
| Qualification/diploma obtained | Obtaining the Habilitation Certificate in the field of doctoral studies in Electrical Engineering<br>Certificate 5365 of 07.10.2021<br>Decision 91/18.10.2021 – obtaining the status of PhD supervisor within the Doctoral School of Electrical Engineering of the Politehnica University of Bucharest, in the field of doctoral studies in Electrical Engineering |
| Name and type of organisation providing education and training | Politehnica University of Bucharest |
| Dates | 21.03.2022 – 26.05.2022 |
| Qualification/diploma obtained | Graduation Certificate - specialization in the training program Social Services Manager<br>Certificate issued by the Ministry of Labour and Social Protection, Series N, No. 0108396 |
| Name and type of organisation providing education and training | NEWINFO COMPANY SRL |
| Dates | 10/2010 → |
| Qualification/diploma obtained | Cambridge Certificate. Level obtained: Council of Europe LEVEL C1 |
| Name and type of organisation providing education and training | University of Cambridge ESOL Examinations |
| Dates | 2009 - 2010 |
| Qualification/diploma obtained | Graduation certificate for employment in teaching, in accordance with the legal regulations in force.<br>Overall average: 9.86. |
| Principal subjects/occupational skills covered | Educational Pedagogy, Pedagogy, Educational Communication, Educational Management |
| Name and type of organisation providing education and training | Politehnica University of Bucharest, Department for Teacher Training. (Postgraduate Studies) |
| Dates | 2000 - 2008 |
| Qualification/diploma obtained | PhD Engineer. Field: Electrical Engineering. |
| Principal subjects/occupational skills covered | PhD thesis title: "Contributions to the recognition of defect shapes in non-destructive magnetic flaw detection".<br>Scientific supervisor: Prof. Dr. Eng. Ioan Florea Hantila.<br>Diploma obtained according to Ministry of Education Order no. 3030/13.01.2009. |
| Name and type of organisation providing education and training | Politehnica University of Bucharest (Doctoral studies)<br>(Romania) |
| Dates | 02/11/2006 - 02/12/2006 |
| Qualification/diploma obtained | Course: Initial Training |
| Name and type of organisation providing education and training | Course: Initial Training<br>(Spain) |
| Dates | 14/11/2006 - 01/12/2006 |
| Qualification/diploma obtained | Advanced course: ERDAS Imagine |


For more information on Europass go to http://europass.cedefop.eu.int
© European Union, 2002-2010 24082010

| | |
|---|---|
| Name and type of organisation providing education and training | European Union Satellite Centre - Spain<br>(Spain) |
| Dates | 29/01/2007 - 02/02/2007 |
| Qualification/diploma obtained | Course: "Interpretation of nuclear facilities using commercial satellite imagery (NFC-1 |
| Name and type of organisation providing education and training | European Union Satellite Centre - Spain<br>(Spain) |
| Dates | 24/09/2007 - 28/09/2007 |
| Qualification/diploma obtained | Advanced course: "Interpretation of nuclear facilities using commercial satellite imagery (NFC-2) |
| Name and type of organisation providing education and training | European Union Satellite Centre - Spain<br>(Spain) |
| Dates | 26/02/2007 - 02/03/2007 |
| Qualification/diploma obtained | Course: "Interpretation of Industrial Facilities using commercial satellite imagery (IND-1)". |
| Name and type of organisation providing education and training | European Union Satellite Centre - Spain<br>(Spain) |
| Dates | 15/10/2007 - 19/10/2007 |
| Qualification/diploma obtained | Advanced course: Interpretation of Industrial Facilities using commercial satellite imagery (IND-2) |
| Name and type of organisation providing education and training | European Union Satellite Centre - Spain<br>(Spain) |
| Dates | 02/06/2003 - 30/06/2003 |
| Qualification/diploma obtained | Graduation Diploma – VISUAL C++ programming. Final exam grade: 10. |
| Principal subjects/occupational skills covered | VISUAL C++ |
| Name and type of organisation providing education and training | PRO MANAGEMENT Professional Training Centre<br>Bucharest (Romania) |
| Dates | 2001 - 2002 |
| Qualification/diploma obtained | Postgraduate Specialization Studies Diploma. Field: Program and Project Management. Average: 10. |
| Name and type of organisation providing education and training | Bucharest Academy of Economic Studies<br>Bucharest (Romania) |
| Dates | 01/2000 - 11/2000 |
| Qualification/diploma obtained | Professional qualification certificate - National Tourist Guide profession |
| Principal subjects/occupational skills covered | Geography of Romania, History of Art, History of Romania, Communication Techniques. |
| Name and type of organisation providing education and training | S.C. O.N.T. Carpati S.A. |
| Dates | 1995 - 2000 |
| Qualification/diploma obtained | Graduate Engineer. Specialization: Electrical and Computer Engineering. Overall graduation average: 8.91. Diploma exam average: 10 |
| Name and type of organisation providing education and training | Politehnica University of Bucharest. Faculty/Department of Engineering Sciences in Foreign Languages (English Branch), Electrical Profile<br>Bucharest (Romania) |
| Dates | 1991 - 1995 |
| Qualification/diploma obtained | Baccalaureate Diploma. Skills: Computer Science. Overall average of study years: 9.45. Overall baccalaureate average: 9.65. |
| Name and type of organisation providing education and training | George Calinescu Theoretical High School – Bucharest, Computer Science Section<br>Bucharest (Romania) |


For more information on Europass go to http://europass.cedefop.eu.int
© European Union, 2002-2010 24082010

## Personal skills and competences

Mother tongue(s)/Foreign language(s)

| Self-assessment | Understanding | | | | Speaking | | | | Writing | |
|---|---|---|---|---|---|---|---|---|---|---|
| *European level (*)* | Listening | | Reading | | Spoken interaction | | Spoken production | | | |
| English | C1 | Proficient user | C1 | Proficient user | C1 | Proficient user | C1 | Proficient user | C1 | Proficient user |
| French | A1 | | A1 | | A1 | | A1 | | A1 | |
| Spanish | A1 | | A1 | | A1 | | A1 | | A1 | |

*(*) Common European Framework of Reference for Languages*

Social skills and competences

Dynamic, communicative, responsible, flexible personality, able to work in a team.

Organisational skills and competences

Vice-chairman of the Local Organizing Committee – International Conference Japmed'06, Bucharest, 27-29 July 2009.

2. Vice-chairman of the Local Organizing Committee - National Symposium of Theoretical Electrical Engineering, 2010 edition
3. Organizing committee for the ATEE 2013, ATEE2015, ATEE2017, ATEE2019, ATEE 2021 conferences
4. Organizing committee for ISFEE 2014, ISFEE 2016, ISFEE2018

Administrative Director, CRRN Gura Ocnitei;

Organization of a painting exhibition at the Sever Galleries in Targoviste, January 2009

Computer skills and competences

use / troubleshooting – advanced level

Technical skills and competences

Areas of scientific expertise:

Modeling of electric and magnetic field problems.
Experience in wireless transmission of electromagnetic energy as well as in the development of numerical methods for solving electromagnetic field problems.

Scientific research activity: over 80 papers, of which:

- 4 course support books published by a CNCSIS-recognized publisher;
- 5 laboratory/application guides published by a CNCSIS-recognized publisher
- 2 specialty books published by a CNCSIS-recognized publisher
- 16 chapters as author with the prestigious international publisher SPRINGER
- 35 articles (8 in journals and 27 in conference proceedings) indexedWOS
- 31 articles indexed in international databases (BDI), CNCSIS B+ category (17 in journals);
- Over 10 articles presented at non-indexed national conferences

- 1 invention patent: Patent Number(s): RO131291-A2, "Device And Method For Determining Static B-H Characteristic Of Ferromagnetic Materials", authors: Ioan Florea HANTILA, Stelian MARINESCU, Viorel STANCIU, Paul Cristian ANDREI, Mihai MARICARU, Marilena STANCULESCU, Derwent Primary Accession Number: 2016-52539F, International Patent Classification: G01N-027/72; G01R-033/12.

List of representative scientific publications: https://orcid.org/0000-0001-7142-5725.

Project Director/Manager:

1. Project Director: National Grants (GNaC 2018 ARUT). Funding contract no. 03.15.10.2018. Project title: Inductive Energy Transfer Applied to Contactless and Human-Intervention-Free Charging of Batteries for Electric Vehicles - TRINEN.
2. Project manager of the Innovation Check research grant 230CI of 01/07/2018, OPTIMIZATION OF THE GEOMETRY OF AN ELECTRIC GENERATOR TO INCREASE ENERGY EFFICIENCY, PN-III-P2-2.1-CI-2018-1343.


For more information on Europass go to http://europass.cedefop.eu.int
© European Union, 2002-2010 24082010

3. Project manager of the Innovation Check research grant 216CI of 25/07/2018, EQUIPMENT FOR WIRELESS CHARGING OF MOBILE PHONES, PN-III-P2-2.1-CI-2018-1275, national competition 2018.

Member of the project team

1. FARCROSS: Facilitating cross-border transmission of electricity through innovation is a research and development project funded by the European Commission under the Horizon2020 programme (Grant Agreement no. 864274), Oct. 2019 – Sept. 2023.
2. Member of the research team - Horizon 2020 Project INTERRFACE - TSO-DSO-Consumer INTERFACE aRchitecture to provide innovative grid services for an efficient power system, Project reference: 824330; Topic: LC-SC3-ES-5-2018-2020; Call for proposals: H2020-LC-SC3-2018-ES-SCC, 2019-2022, 95000 Euro, link: http://www.interrface.eu/
3. Member of the implementation team, PROJECT POCU ID131213 - Synergy between academia and industry to meet the current and future needs of the labour market in the electrical, energy and materials science fields (PRACTICE), budget: 4,058,248 lei, implementation period 01.10.2020-30.09.2022
4. Member of the research team - Project PED696/2022 - Remote control and telemonitoring of an outdoor vacuum-switching high-voltage circuit breaker for the railway system. Total value 694,795.00 lei, (2022-2024).
5. Member of the research team, PN-III-P2-2.1-PTE-2019-0423, Innovative electric motor with rare-earth-free permanent magnets for light electric vehicles (EMLEV), Budget: 1,189,000 lei, 2019-2021.
6. 30/SGU/NC/I/ "Improving opportunities for participation in quality education within the Faculty of Electrical Engineering - TOPICFIE", grant funded under the University Grants Scheme, carried out as part of the Secondary Education Project - ROSE, Ministry of National Education, 01.11.2017-17.10.2019
7. THERMAL SUPPORT FOR THE DETECTION AND LOCALIZATION OF BREAST CANCER - TERMOCAN 30PTE of 06/10/2016 (PN-III-P2-2.1-PTE-2016-0029).
8. Member of the project team: Electromagnetic methods for improving winemaking processes - PN II: PARTNERSHIPS IN PRIORITY AREAS, ID: PN-II-PT-PCCA-2013-4-2225 - period: 2014-2016 - contract 170/2014, MEN-UEFISCDI
9. Member of the project team: Localization and size evaluation of breast tumors by solving the inverse thermal field problem (INVTERM) (AMCSIT CEEX project 125/2006, 2006-2008).
10. Member of the project team: Controlling solidification structure formation by means of the electromagnetic field (SOLID) (AMCSIT CEEX project 300/2006, 2006-2008).
11. Member (co-author), Reconstruction of defect shapes by electromagnetic methods (MAGNED), AMCSIT CEEX Project 79/2005, 2005-2007.
12. Member of the project team: Equipment for drying walls in a high-frequency electromagnetic field (ELMDRY), PN-II-P2-2.1-CI-2017-0358.
13. Member of the project team: UPB-GEX, contract 73/Sept 2016, code 439 Electromagnetic Stimuli for Living Tissues, 2006-2007.

Other skills and competences

Reviewer for journals and conferences

*ISI-indexed journal*: Revue Roumaine des Sciences Techniques – Série Electrotechnique et Energétique, Editura Academiei Române, ISSN 0035-4066

*BDI-indexed journal*: The Scientific Bulletin of Electrical Engineering Faculty, CNCSIS Category B+, CNCSIS code 830, ISBN 1843-6188 (BDI:IndexCopernicus)

*MDPI journals*

Additional

*ISI-indexed conferences*: ATEE2015, ATEE 2017, ATEE2019, ATEE 2021, ATEE 2023, ISFEE 2018, EEEIC / I&CPS Europe 2019.

Member of the Romanian Association of Electrical and Electronics Engineers

Expert Evaluator for European Programmes

Driving licence

B

Representative works:

*List of representative scientific publications*: https://orcid.org/0000-0001-7142-5725.


For more information on Europass go to http://europass.cedefop.eu.int
© European Union, 2002-2010 24082010

1. Andrei, P.C. , Stanculescu Marilena, Andrei, H., Caciula, I., Diaconu, E., Bizon N., Mazare A.G., Ionescu L.M., Gaiceanu, M., *Comparative and Predictive Analysis of Electrical Consumption during Pre- and Pandemic Periods: Case Study for Romanian Universities*, Source: SUSTAINABILITY, Volume: 14, Issue: 18, Article Number: 11346, DOI: 10.3390/su141811346, Published: SEP 2022, (22 pages), Accession Number: WOS:000856632500001, eISSN: 2071-1050, IF=3.889.
2. Stanculescu M., Andrei P.C., Andrei H., Deleanu S., Bobaru L. (2021) *Numerical Assessment of Electromagnetic Energy and Forces in Non-destructive Measurement Devices*. In: Mahdavi Tabatabaei N., Bizon N. (eds) Numerical Methods for Energy Applications. Power Systems. Springer, Cham. https://doi.org/10.1007/978-3-030-62191-9_21, pp. 554-577 (24 pages), ISBN: 978-3-030-62191-9, indexed in SCOPUS.
3. Stanculescu M., Deleanu S., Andrei P.C., Andrei H., Bobaru L., Iordache M. (2021) *Theoretical Approaches of Finite Elements Method (FEM)*. In: Mahdavi Tabatabaei N., Bizon N. (eds) Numerical Methods for Energy Applications. Power Systems. Springer, Cham. https://doi.org/10.1007/978-3-030-62191-9_3, pp. 61-92 (32 pages), ISBN: 978-3-030-62191-9, indexed in SCOPUS.
4. Niculae Dragos Marin, Stanculescu Marilena, Deleanu Sorin, Iordache Mihai, Bobaru, Lavinia, *Wireless Power Transfer Systems Optimization Using Multiple Magnetic Couplings*, Volume 10, Issue 20, Article Number 2463, DOI 10.3390/electronics10202463, Published OCT 2021, Indexed 2021-11-13, Document Type Article, (15 pages), WOS:000715322800001, eISSN 2079-9292. Journal: Electronics, IF=2.69.
5. Ioan Florea Hantila, Mihai Maricaru, Marilena Stanculescu, George Marian Vasilescu, *Method For Analyzing Three-Phase Networks With Nonlinear Resistive Elements*, Revue Roumaine Des Sciences Techniques-Serie Electrotechnique Et Energetique, Volume: 64, Issue: 2, Pages: 103-106 (4 pages), Published: APR-JUN 2019, Accession Number: WOS:000479328200001, ISSN: 0035-4066, IF=0.670
6. Andrei H., Gaiceanu M., Stanculescu M., Arama I.N., Andrei P.C. (2019) *Power Systems Connectivity and Resiliency*. In: Mahdavi Tabatabaei N., Najafi Ravadanegh S., Bizon N. (eds) Power Systems Resilience. Power Systems. Springer, Cham, https://doi.org/10.1007/978-3-319-94442-5_2, Print ISBN 978-3-319-94441-8, Online ISBN 978-3-19-94442-5, eBook Packages Energy, pp. 45-79 (35 pages), WOS:000456874500004, indexed in SCOPUS.
7. Marilena Stanculescu, Mihai Iordache, Dragos Niculae, Lavinia Bobaru, Victor Bucata, *Algorithm for Computing S Parameters and Their Use for Studying Efficiency of Electromagnetic Energy Wireless Transfer System*s, Revue Roumaine Des Sciences Techniques-Serie Electrotechnique Et Energetique, Volume: 63, Issue: 2, Pages: 138-144 (6 pages), Published: APR-JUN *2018*, IF=1.114 (2017), Q3 category, Accession Number: WOS:000438662400005, ISSN: 0035-4066, indexed in SCOPUS.
8. Paul Cristian Andrei, Mihai Maricaru, Stelian Marinescu, Marilena STANCULESCU, Ioan Florea Hantila, *An Efficient Procedure to Assess The Static Magnetization Relationship*, Revue Roumaine Des Sciences Techniques-Serie Electrotechnique Et Energetique, Vol. 61, Issue 2, pp. 101-105 (4 pages), Apr-Jun *2016*, ISI Thomson Reuters WOS:000381238000001, ISSN: 0035-4066, IF = 1.114, indexed in SCOPUS.
9. Marilena STANCULESCU, M. Marifcaru, V. Stefan-Minculete, S. Marinescu, I.F. Hantila, *Analytical Solution for Eddy Current Problem, Using Space Eigenfunctions Expansion*, Revue Roumaine Des Sciences Techniques-Serie Electrotechnique Et Energetique, Volume 58, Issue 2, Pages 123-134 (10 pages), Published Apr-Jun 2013, ISI Thomson Reuters WOS:000320488100002, ISSN: 0035-4066, IF = 1.114, indexed in SCOPUS.
10. Marilena STANCULESCU, M. Maricaru, I.F. Hantila, S. Marinescu, Livia Bandici, *An Iterative Finite Element - Boundary Element Method For Efficient Magnetic Field Computation In Transformers*, Revue Roumaine Des Sciences Techniques-Serie Electrotechnique Et Energetique, Volume: 56, Issue: 3, Pages: 267-276 (10 pages), Published: JUL-SEP 2011, ISI Thomson Reuters WOS:000298453900003, ISSN: 0035-4066, IF = 1.114, indexed in SCOPUS.


© European Union, 2002-2010 24082010